

60-DAY DATA ENGINEER ROADMAP

Master SQL, Data Modeling, Databricks & More

For Senior Data Engineer / Data Platform Engineer Roles in UAE

Goal: Land a Senior Data Engineer / Data Platform Engineer role in UAE product companies through focused, project-based learning of the highest-ROI technical skills.

Duration	Daily Commitment	Phases	Projects
60 Days	4 Hours	5 Phases	5+ Repos

PHASE 1: SQL + DBMS (Days 1–14)

Why This Matters?

Highest interview ROI. Most commonly asked in data engineer interviews across UAE tech companies.

SQL Topics

Basic Level: Joins, Aggregations, Group By, Having

Intermediate Level: CTEs, Window Functions, Ranking

Advanced Level: Cohort Analysis, Funnel Analysis, Recursive CTEs

DBMS Concepts

- ACID Properties
- Isolation Levels
- Indexing & Query Optimization
- Query Execution Plans
- Partitioning

Deliverables

- ✓ 75 solved SQL questions with explanations
- ✓ 10-page DBMS comprehensive notes

Real Interview Questions

1. Explain ACID properties with examples. Why is each important?
2. Write a query to find top 5 customers by revenue using window functions
3. How would you optimize a slow query? Walk through your approach

PHASE 2: Data Modeling + Warehousing (Days 15–28)

Data Modeling Concepts

- Fact Tables & Dimension Tables
- Star Schema Design
- Snowflake Schema
- SCD Type 1 & Type 2 implementation

Real-World Case Studies

Amazon Analytics: Orders, Customers, Products

Netflix Analytics: Users, Watch History, Content

Uber Analytics: Trips, Drivers, Riders

Data Warehousing Approaches

- Kimball Methodology
- Data Vault Basics
- Lakehouse Architecture
- Medallion (Bronze/Silver/Gold)

Deliverables

- ✓ 3 architecture diagrams (Amazon, Netflix, Uber analytics)
- ✓ Data model documentation for each case

Real Interview Questions

1. Design Amazon's data warehouse. Explain your fact and dimension tables
2. What is SCD Type 2? When would you use it? Show implementation
3. Compare Star Schema vs Data Vault. Trade-offs?

PHASE 3: Databricks + Delta Lake + PySpark (Days 29–40)

Delta Lake Mastery

- Delta Tables & Time Travel
- Merge & Upserts
- Optimize & Z-Order

PySpark Core Concepts

- Partitions & Shuffles
- Broadcast Joins
- Lazy Evaluation
- Adaptive Query Execution (AQE)

Learning Resource

Use: Databricks Partner Academy (Free access)

Capstone Project

Retail Analytics Lakehouse Pipeline:

Raw Data → Bronze (Raw ingestion) → Silver (Cleaned & validated) → Gold (Analytics-ready)

Real Interview Questions

1. You have a Spark job running in 30 mins. How would you optimize it to 5 mins?
2. Explain Delta Lake's ACID guarantees. How does it handle conflicts?
3. When would you use broadcast join vs sort merge join?

PHASE 4: Kafka + Airflow + System Design (Days 41–50)

Kafka Streaming

- Producers & Consumers
- Partitions & Consumer Groups
- Kafka Connect

Apache Airflow Orchestration

- DAG Design & best practices
- Scheduling & monitoring
- Retry logic & backfills

System Design Projects

- Clickstream Analytics Platform
- Customer 360 Platform
- Fraud Detection Pipeline

Deliverables

- ✓ All system designs in Draw.io / Lucidchart
- ✓ Complete architecture documentation

Real Interview Questions

1. Design a Kafka-based real-time analytics pipeline. Handle millions of events/day
2. How would you handle late-arriving data in Kafka?
3. Design an Airflow DAG for daily data warehouse refresh

PHASE 5: GenAI + RAG (Days 51–60)

Learn

- LLM Fundamentals
- Embeddings & Vector Databases
- Prompt Engineering
- RAG Architecture

Build: Data Engineering Copilot

Features:

- Generate SQL from natural language
- Explain database schemas
- Generate data documentation
- Suggest data models

Tech Stack: Python + Streamlit + OpenAI/Claude + ChromaDB

Real Interview Questions

1. Explain RAG architecture. When would you use it vs fine-tuning?
2. How would you evaluate RAG system quality?
3. Build a simple RAG system and explain trade-offs

TOOLS & ENVIRONMENT SETUP (Before Day 1)

Essential Setup

Tool	Purpose	Installation
PostgreSQL	Learn SQL	Docker: docker run --name postgres -e POSTGRES_PASSWORD=pwd -p 5432:54
Python 3.9+	PySpark, scripting	brew install python3 (Mac) or apt install python3 (Linux)
PySpark	Spark development	pip install pyspark
Git & GitHub	Version control	Install git, create GitHub account
VS Code	Code editor	https://code.visualstudio.com/
Databricks Account	Free tier access	https://databricks.com/try-databricks
Draw.io	Architecture diagrams	https://app.diagrams.net (free web version)

DAILY SCHEDULE (4 Hours Daily)

Hour	Focus	Activities
Hour 1 (Morning)	LEARNING	Databricks Academy, LinkedIn Learning
Hour 2	HANDS-ON	SQL, Spark, Python coding
Hour 3	ARCHITECTURE	Data modeling, System design
Hour 4 (Wrap-up)	PROJECT	GitHub commits, Documentation

RECOMMENDED FREE RESOURCES

SQL & DBMS

- LeetCode Database (75 SQL problems): <https://leetcode.com/problemset/database/>
- SQLZoo (Interactive SQL tutorial): <https://sqlzoo.net/>
- Mode Analytics SQL Tutorial: <https://mode.com/sql-tutorial/>

Data Warehousing & Modeling

- Kimball Group Resources (Free): <https://www.kimballgroup.com/>
- YouTube: Salma Dardouri (Data Modeling): <https://youtube.com/@SalmaDardouri>

Databricks & PySpark

- Databricks Academy (Free): <https://academy.databricks.com/>
- YouTube: Data Engineering Simplified: <https://youtube.com/@DataEngineeringSimplified>

Kafka & System Design

- Kafka Official Documentation: <https://kafka.apache.org/documentation/>
- System Design Primer: <https://github.com/donnemartin/system-design-primer>
- YouTube: Fireship (System Design): <https://youtube.com/@Fireship>

GenAI & RAG

- LangChain Documentation: <https://python.langchain.com/>
- OpenAI API Docs: <https://platform.openai.com/docs>
- YouTube: ArjanCodes (AI Architecture): <https://youtube.com/@ArjanCodes>

WEEKLY MILESTONES & CHECKPOINTS

Week	Phase	Key Milestones	GitHub Deliverable
Week 1	SQL Basics	Complete 25 SQL Q Learn 5 DBMS concepts	Create `sql-mastery` repo created
Week 2	DBMS Advanced	Solve 50 Q Write DBMS notes	Push DBMS guide & SQL solutions
Week 3	Data Modeling	Design 3 schemas Create 5 ERDs	Create `data-modeling` repo
Week 4	Data Warehouse	Complete warehouse design Document	Push Kimball/Vault guides
Week 5	PySpark	Build 3 Spark jobs Optimize queries	Create `pyspark-optimization` repo
Week 6	Kafka & Airflow	Design 3 architectures Code DAGs	`system-design` repo with diagrams
Week 7	GenAI & RAG	Build copilot MVP Document architecture	Create `data-copilot` repo (demo)
Week 8	Portfolio & Interview	Polish all repos Mock interviews	Final portfolio review

COMMON MISTAKES TO AVOID

- Mistake #1: Memorizing Instead of Understanding

Don't memorize 75 SQL solutions. Understand the patterns, apply them to new problems.

■ **Mistake #2: Not Building Projects**

Reading about Spark $\neq$ Using Spark. Code every day. Build real projects.

■ **Mistake #3: Weak GitHub Portfolio**

Recruiters check GitHub first. Commit daily. Write clear READMEs. Use proper documentation.

■ **Mistake #4: Skipping System Design**

Many engineers skip this. System design questions are critical in interviews. Practice it.

■ **Mistake #5: Getting Distracted by Advanced Topics**

Ignore ML, fine-tuning, reinforcement learning. Focus on the 5 phases. Stay disciplined.

■ **Mistake #6: Not Tracking Progress**

Keep a simple log. What did you learn today? What's your weekly checkpoint score?

PORTFOLIO CHECKLIST: GitHub Repositories to Create

Repository	Description	Key Files	Status
sql-mastery	75 SQL problems + solutions + explanations	README.md, 75 .sql files, SOLUTIONS.md	■
data-warehouse-designs	Amazon, Netflix, Uber warehouse ER diagrams	ER diagrams, design docs, SQL schemas	■
dbms-fundamentals	ACID, Indexing, Query Plans, Partitioning	Diagram notes, diagrams, code examples	■
databricks-pyspark-optimization	Spark optimization patterns & real examples	Spark optimization guide, benchmarks	■
system-design-archive	Kafka pipelines, Airflow DAGs, additional designs	Architecture diagrams, code	■
data-copilot	AI-powered data engineering assistant (RAG)	Python RAG app, deployment guide	■
interview-preparation	All interview Q&A, STAR stories, practice interview	Q&A document, video links, notes	■

SIMPLE DAILY PROGRESS TRACKER

Use this template to track daily progress:

Date: _____ Phase: _____ Week: _____

Hour 1 (Learning): [Topic studied] | Minutes spent: ____ | Key takeaway: _____
Hour 2 (Hands-on): [What you coded] | Minutes spent: ____ | Issue/Success: _____
Hour 3 (Architecture): [Design work] | Minutes spent: ____ | Key learning: _____
Hour 4 (Project): [GitHub work] | Commits: ____ | What you pushed: _____

- Weekly Checkpoint: (Every Friday)
- Completed 25% of phase goals
 - All code pushed to GitHub
 - Created 2+ documentation files
 - Ready for next week

RESUME SKILLS AFTER 60 DAYS

Data Engineering

- SQL (Advanced: Window functions, CTEs, recursive)
- PySpark & DataFrame API
- Delta Lake & Time Travel
- Databricks Workspace

- Data Warehousing Architecture
- Data Modeling (Star Schema, Kimball, Data Vault)

Cloud & Infrastructure

- AWS (Glue, Athena, Redshift, Lambda)
- Lakehouse Architecture
- CDC (Change Data Capture)

Big Data & Streaming

- Apache Kafka
- Apache Airflow
- Distributed Systems Design

AI/ML Integration

- RAG Architecture
- Vector Databases
- Prompt Engineering
- AI-powered Data Applications

FOCUS: What NOT to Study (Skip Until After Job Switch)

These topics are unlikely to be asked in Senior Data Engineer interviews for UAE product companies in the next 60 days. Skip them for now.

- Deep Learning (TensorFlow, PyTorch)
- Fine-Tuning Models
- MLOps
- Reinforcement Learning
- Multi-Agent Research

SUCCESS METRICS: By Day 60, You Should Master

1. Design Amazon's data warehouse with star schema

2. Explain SCD Type 2 with real examples
3. Optimize a slow Spark job (partitions, shuffles, broadcasts)
4. Design a Kafka-based real-time analytics pipeline
5. Explain Delta Lake internals & ACID guarantees
6. Compare Redshift vs Athena vs Snowflake
7. Design a complete Lakehouse on AWS
8. Explain RAG architecture & use cases
9. Build and demo your Data Engineering Copilot

If you can confidently answer these 9 questions, you'll be positioned significantly above many candidates. You'll be ready for Senior Data Engineer interviews.

BONUS: What's Next After 60 Days

Advanced Topics (Start After Job Switch)

- Real-time ML Feature Stores
- Data Quality & Observability (Great Expectations, Soda)
- Advanced Spark Tuning & Cluster Management
- Kubernetes & Containerization for Data Pipelines
- dbt (Data Build Tool) for transformation
- Advanced Kafka Patterns (Event Sourcing, CQRS)

Open Source Contributions

After securing the job, contribute to:

- Apache Spark
- Delta Lake
- LangChain
- Databricks open source projects

Networking & Community

- Join local UAE tech meetups (Dubaitech, Abu Dhabi Dev Community)
- Follow data engineers on LinkedIn in UAE (Flipkart, Noon, Careem, Dubizzle)
- Engage in online communities (Reddit r/dataengineering, Discord servers)
- Attend tech conferences (Infosec, Dubai DevOps)

Remember: This is a marathon, not a sprint. You have 60 focused days to master the most important concepts. Stay consistent, build in public, and track your progress. By Day 60, you'll have built a portfolio that demonstrates you're ready for a Senior Data Engineer role in any UAE tech company.

Good luck! ■